

Chapter 1, Problem 22P

Problem

A lightning bolt strikes an airplane with 40 kA for 1.7 ms. How many coulombs of charge are deposited on the plane?

Step-by-step solution

Step 1 of 1

The current strikes on an airplane $i = 40 \text{ kA}$

The time that the current strikes on the airplane $= 1.7 \text{ ms}$

Calculate the charge deposited on the airplane.

$$\begin{aligned} q &= it \\ &= (40 \text{ kA})(1.7 \text{ ms}) \\ &= (40 \times 10^3)(1.7 \times 10^{-3}) \\ &= 68 \text{ C} \end{aligned}$$

Therefore, the charge q deposited on the airplane is 68 C.